

Deployment Guide

Local, Docker & Cloud Deployment Reference

FinClaw — AI-Assisted Paper Trading Dashboard

1. Project Overview

This guide covers all deployment paths for FinClaw: local development setup, Docker Compose for containerised local deployment, and cloud platform options for staging/production.

2. Prerequisites

- Python 3.11+
- Node.js / npm (optional, for frontend dev server)
- Docker + Docker Compose (for containerised deployment)
- Git
- Alpaca paper trading account (optional)

3. Local Setup — Backend

```
cd backend
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
copy .env.example .env
# Edit .env with your values
python seed.py
uvicorn app.main:app --reload --host 127.0.0.1 --port 8000
```

4. Local Setup — Frontend

```
cd frontend
python -m http.server 3000
# OR
npm run start
```

Open: <http://127.0.0.1:3000/login.html>

5. Docker Compose Deployment

```
cd "ai trading"
docker compose up --build
```

Service	Port	Description
backend	8000	FastAPI + Uvicorn
frontend	3000	Static file server

```
# Stop
docker compose down
# Stop and remove volumes
docker compose down -v
```

6. Environment Variables

Variable	Default	Description
APP_NAME	FinClaw API	Application name
APP_ENV	development	Environment tag

DATABASE_URL	sqlite:///./finclaw.db	DB connection string
CORS_ORIGINS	*	Allowed CORS origins
ALPACA_API_KEY	(empty)	Alpaca key (optional)
ALPACA_SECRET_KEY	(empty)	Alpaca secret (optional)
ALPACA_PAPER	true	Use paper trading
DEFAULT_CASH_LIMIT	500	Max daily spend
DEFAULT_MAX_ORDER_NOTIONAL	100	Max single order USD
DEFAULT_MAX_DAILY_TRADES	3	Max trades per day
DEFAULT_APPROVED_TICKERS	AAPL,MSFT,NVDA,GOOGL,AMZN,FB,DIS	Allowed tickers

7. Cloud Deployment Options

Platform	Method
AWS App Runner	Deploy backend/ as Python web service; run <code>uvicorn app.main:app --host 0.0.0.0 --port 8000</code>
Render / Railway	Same as App Runner; set env vars in dashboard
Fly.io	fly launch from backend/; set secrets via fly secrets set
Netlify / Vercel	Serve frontend/ as static files; backend must run separately

8. Production Checklist

- Replace SQLite with PostgreSQL — set DATABASE_URL accordingly
- Restrict CORS_ORIGINS to your frontend domain
- Set real ALPACA_API_KEY and ALPACA_SECRET_KEY as environment secrets only
- Put frontend behind a CDN (CloudFront, Cloudflare)
- Add JWT/session auth before handling real users
- Enable HTTPS on all endpoints
- Set APP_ENV=production

9. Health Check

```
GET http://127.0.0.1:8000/health
Response: {"ok": true, "app": "FinClaw API", "env": "development"}
```

10. URLs Reference

Resource	URL
Frontend	<code>http://127.0.0.1:3000/login.html</code>
Backend API	<code>http://127.0.0.1:8000</code>
Swagger Docs	<code>http://127.0.0.1:8000/docs</code>

Health Check	http://127.0.0.1:8000/health
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